

Solving Optimal Stopping Problems with Reinforcement Learning

MMathSci Thesis Presentation

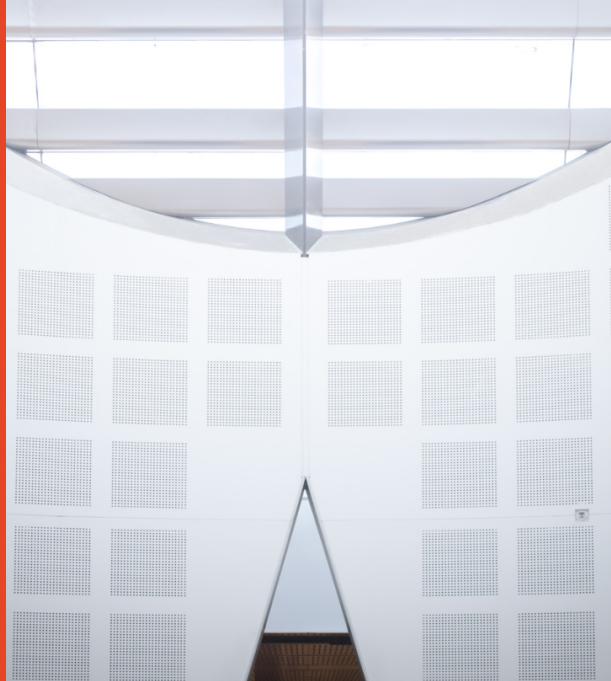
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THE UNIVERSITY OF
SYDNEY



What is Optimal Stopping?

In optimal stopping, we find the best time τ to **perform an action**, maximising:

$$\sup_{\tau} \mathbb{E}[e^{-\alpha\tau} X_{\tau}]$$

- ▶ X : **Stochastic process** - specifies reward over time.
- ▶ α : **Discounting rate** - incentivises stopping.
- ▶ τ : **Stopping time** - represents time to act.

X is a risky process, but at time τ we lock in the reward.

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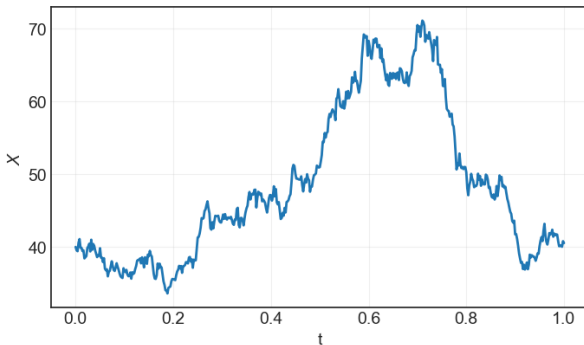
There are several real-world problems involving optimal stopping:

- ▶ **Mathematical finance:** Asset pricing, portfolio optimisation.
 - ▶ X : stock price, τ : time to sell.
- ▶ **Engineering:** Sequential detection, signal processing.
 - ▶ X : observed signal, τ : time to raise alarm.

What is Optimal Stopping?

Suppose X is a stock price following $dX_t = b(X_t)dt + \sigma(X_t)dW_t$.

- ▶ You buy at $X_0 = 40$. What is the optimal time to sell?



What if we don't know the underlying dynamics of X ?

What is Reinforcement Learning?

Reinforcement learning develops effective strategies by **learning from itself**.

- ▶ Makes **no assumptions** on model parameters.
- ▶ Updates strategy after each observation.

Reinforcement learning strikes a balance between:

- ▶ **Exploitation**: Maximise rewards based on current model.
- ▶ **Exploration**: Try new actions to improve model accuracy.

Applying Reinforcement Learning to Optimal Stopping

Optimal stopping

- ▶ Precise mathematical formulation.
- ▶ Clear notion of optimality.
- ▶ Requires specified model.

Reinforcement learning

- ▶ Learns policies from data.
- ▶ Model-free algorithm.
- ▶ Scales to complex settings.

In this thesis we develop a reinforcement learning framework for several classes of optimal stopping problems.

Applying Reinforcement Learning to Optimal Stopping

Classical optimal stopping: Pure stopping time τ .

$$\sup_{\tau} \mathbb{E}[e^{-\alpha\tau} g(X_{\tau})]$$

- ▶ Example: Stop when $X > 50$.

Randomised optimal stopping: Stopping intensity π_t .

$$\sup_{\pi} \mathbb{E} \left[\int_0^T e^{-\alpha t} g(X_t) \pi_t p_t dt + e^{-\alpha T} g(X_T) p_T \right]$$

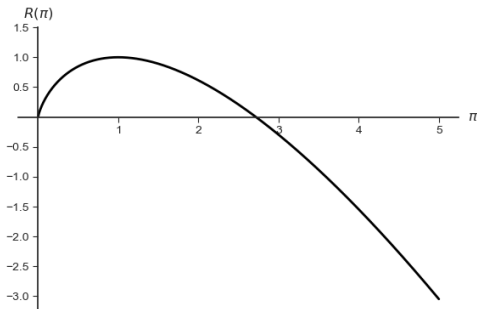
- ▶ p_t : Probability of survival to time t .
- ▶ $\pi_t p_t dt$: Probability of stopping over $[t, t + dt]$.
- ▶ Example: Stop with higher probability when $X > 50$.

Applying Reinforcement Learning to Optimal Stopping

Problem: These problems are identical!

► Optimal intensity: $\pi_t = \infty$ when $t \geq \tau^*$, $\pi_t = 0$ otherwise.

Solution: Introduce entropy $R(\pi) = \pi - \pi \log \pi$ to encourage exploration.



► $R(0) = 0$, $R(\pi) \rightarrow -\infty$ as $\pi \rightarrow \infty$.

► Maximised at $\pi = 1$.

Applying Reinforcement Learning to Optimal Stopping

This gives us the **entropy-regularised objective**:

$$\sup_{\pi} \mathbb{E} \left[\int_0^T e^{-\alpha t} (g(X_t) \pi_t p_t + \lambda R(\pi_t) p_t) dt + e^{-\alpha T} g(X_T) p_T \right]$$

- ▶ λ : Exploration parameter - larger λ promotes exploration.

How do we solve the entropy-regularised problem?

Results

Recall the entropy-regularised objective:

$$V^\pi(t, x) = \mathbb{E} \left[\int_t^T e^{-\alpha(s-t)} (g(X_s) \pi_s p_s + \lambda R(\pi_s) p_s) ds + e^{-\alpha(T-t)} g(X_T) p_T \right]$$

Theorem (HJB Equation and Optimal Policy)

For a fixed intensity π , the value $V^\pi(t, x)$ solves the PDE

$$(\partial_t + \mathcal{L})V^\pi(t, x) - \alpha V^\pi(t, x) + (g(x) - V^\pi(t, x))\pi(t, x) + \lambda R(\pi(t, x)) = 0,$$

with terminal condition $V^\pi(T, x) = g(x)$.

Taking the supremum over $\pi \geq 0$ (with $V^ = \sup_{\pi} V^\pi$) yields the maximiser*

$$\pi^*(t, x) = \exp \left(-\frac{V^*(t, x) - g(x)}{\lambda} \right).$$

Results

Recall the entropy-regularised objective:

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Theorem (Monotone Policy Iteration)

Define (π_n, V^{π_n}) iteratively by:

- ▶ **Policy evaluation:** given π_n , solve the linear PDE for V^{π_n} .
- ▶ **Policy update:** set $\pi_{n+1}(t, x) = \exp\left(-\frac{V^{\pi_n}(t, x) - g(x)}{\lambda}\right)$.

Then:

$$V^{\pi_n} \uparrow V^* \quad \text{and} \quad \pi_n \downarrow \pi^* \quad \text{as } n \rightarrow \infty.$$

Using Reinforcement Learning in Practice

In practice, we do not solve the PDE exactly. We instead approximate V with a parametric model V_θ and train it using regression.

$$\pi_1 \longrightarrow V_\theta^{\pi_1} \longrightarrow \pi_2 \longrightarrow V_\theta^{\pi_2} \longrightarrow \dots \longrightarrow \pi^* \longrightarrow V_\theta^*$$

For our numerical examples, we use a fully connected neural network to learn the model from simulated data alone and compute the relative error to the theoretical value.

Numerical Example - American Put

We use the policy iteration algorithm on the classical American put

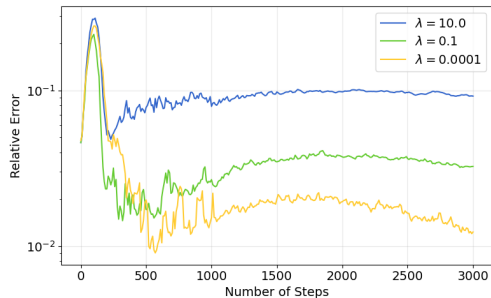
$$g(X) = (K - X)^+, \quad dX_t = X_t(r dt + \sigma dW_t)$$

with parameters

$$X_0 = K = 40, \quad r = 0.06, \quad \sigma = 0.40, \quad T = 1.$$

Interpretation: The holder has the right to sell the asset X to the issuer at any time for the agreed strike price K .

Numerical Example - American Put



Adjusting λ gives us a tradeoff:

- ▶ Larger λ accelerates convergence.
- ▶ Smaller λ reduces bias.

Summary & Further Extensions

The randomised method provides a consistent way to solve optimal stopping problems when the underlying dynamics are unknown and must be learned from data.

The intensity-based viewpoint provides a scaffold for several extensions:

- ▶ **Double stopping:** Problems that require multiple exercise times.
- ▶ **Time-inconsistent stopping:** Discounting is no longer exponential.
- ▶ **Adaptive exploration:** Allow λ to vary over training.

Double Stopping Problem

In the double stopping problem choose two stopping times (τ_1, τ_2) to maximise

$$v(S) = \operatorname{ess\,sup}_{\tau_1, \tau_2 \in \mathcal{T}_{[S, T]}} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S].$$

Theorem (Reduction Principle)

For each θ , define

$$u_1(\theta) = \operatorname{ess\,sup}_{\tau_1 \geq \theta} \mathbb{E}[\psi(\tau_1, \theta) \mid \mathcal{F}_\theta], \quad u_2(\theta) = \operatorname{ess\,sup}_{\tau_2 \geq \theta} \mathbb{E}[\psi(\theta, \tau_2) \mid \mathcal{F}_\theta],$$

and set $\phi(\theta) = \max\{u_1(\theta), u_2(\theta)\}$. Then the single-stopping value

$$u(S) = \operatorname{ess\,sup}_{\theta \geq S} \mathbb{E}[\phi(\theta) \mid \mathcal{F}_S]$$

satisfies $v(S) = u(S)$ almost surely.

References

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